

TASK 2

OBJECTIVE: To Built a python program that calculates a user's Body Mass Index (BMI) and classifies it into health categories. Beginners build a command-line tool; advanced builds a full GUI application with data persistence and trend visualization.

BMI Calculator

```
Print ("==== BMI Calculator ====")
```

```
try:
```

```
    # Get input from user
```

```
    weight = float (input ("Enter your weight in kg: "))
```

```
    height = float (input ("Enter your height in meters: "))
```

```
# Validate input
```

```
    if weight <= 0 or height <= 0:
```

```
        print ("Error: Weight and height must be positive values.")
```

```
    else:
```

```
        # Calculate BMI
```

```
        B mi = weight / (height ** 2)
```

```
# Classify BMI
```

```
    if b mi < 18.5:
```

```
        category = "Underweight"
```

```
    e lif bmi < 25:
```

```
        category = "Normal"
```

```
    elif bmi < 30:
```

```
        category = "Overweight"

    else:

        category = "Obese"

# Display result

    Print ("\n===== Result =====")

    Print ("BMI:", round (bmi, 2))

    Print ("Category:", category)

except Value Error:

    print ("Error: Please enter numeric values only.")
```

- ❖ Mainly ..., this Covers
 - ✓ Weight input in KG
 - ✓ Height input in Meters
 - ✓ BMI Formula: $\text{weight} / (\text{height}^2)$
 - ✓ Underweight
 - ✓ Normal
 - ✓ Overweight
 - ✓ Obese
 - ✓ BMI rounded to 2 decimal places
 - ✓ Rejects negative/zero values
 - ✓ Handles non-numeric input

EXAMPLE;

===== BMI Calculator =====

Enter your weight in kg: 60

Enter your height in meters: 1.65

===== Result =====

BMI: 22.04

Category: Normal

OUTPUT:

BMI Calculator, if enter:

Enter your weight in kg: 60

Enter your height in meters: 1.65

The Output will be:

===== BMI Calculator =====

===== Result =====

BMI: 22.04

Category: Normal

- **BMI=22.04**
- **Category=Normal**